

COMPLEMENTARITY AND CRAFT-TRADITION REALISM

Draft

Niels Bohr said late in his life that "... no man who is called a philosopher really understands what one means by the complementarity description."¹ A number of philosophers suspect that something similar could be said about MacIntyre's craft/tradition model of rationality. It is to be hoped that rubbing the two impenetrables together may produce a spark of insight. This paper is an attempt to explore the resources of the craft/tradition model for accounting for some epistemically puzzling features of physical reality at the quantum level. It is possible that the result may render the physics more intelligible, but the primary purpose is to submit the craft/tradition model of rationality to the test of the real world, a criterion which any realist account of rationality must meet. The craft/ tradition model purports to be a realist account, but a fundamentally different one from others which have held center stage in modern realist/antirealist debate.

One chief difference between craft/ tradition realism and rival versions of realism lies in their accounts of the status of extramental reality. It is just on this point that the findings of contemporary physics bear on the contest among realisms and between them and varieties of antirealism. It is no news that certain puzzling features of physical reality at the quantum level seem to provide support for instrumentalism. But those features bear on the debate among varieties of realism as well. In fact, the implications of complementarity for the traditional realist doctrine of the mind-independence of reality suggest that the rationality of craft/tradition is a

1. Quoted in Pais, *Niels Borh's Times* (Oxford University Press, 1991), from a 1962 interview with T.S. Kuhn, *et al.*

promising direction for grounding a realist epistemology of science. In defense of this thesis, some features of physical reality which challenge the doctrine of mind-independence and gave birth to Bohr's notion of complementarity will be rehearsed. In light of the physical reality, a Kuhnian version of instrumentalism will be presented as superior to mind independence realism as an account of scientific knowledge. Craft/tradition realism will be presented as overcoming instrumentalism while incorporating instrumentalist resources for overcoming mind-independence realism. Finally, the relationship of interdependence between the mind and extramental reality suggested by the craft/tradition model will be shown to harmonize with the like relationship suggested by the complementarity thesis.

Complementarity

Complementarity seems a difficult notion to grasp richly, for at least two sorts of reasons. First, the concept is a precise response to a particular set of problems emerging from physical reality. (The notion may prove useful elsewhere, as Bohr thought, but its *raison d'être* is in physics.) So to appreciate the contours and contributions of complementarity one would need to grasp in pretty rich detail enough physics to deeply see into the nature of the conceptual problem engendered by the physical reality and addressed by the solution of complementarity.

Second, complementarity is a notion which seems to involve something of a global change of perspective, a gestalt shift, of sorts. No doubt Einstein and many others had the first sort of difficulty mastered, but saw complementarity

as something of a magic wand trick, a "pillow" on which to rest from difficulty.² So he could describe it, but couldn't capture the appeal of the complementarist picture. It was during a four week skiing vacation that the whole picture pretty well unfolded for Bohr, marking it as the kind of conceptual transformation that Thomas Kuhn describes as typical of the emergence of a new scientific paradigm, and the kind of experience with which Plato was deeply familiar, and to which he referred when he had Socrates remark "Grasp it correctly as a whole, and it will seem perfectly plain to you." Given these sorts of experiences, we should take Bohr very seriously, as the authoritative interpreter of a world view we probably have not yet grasped, when he said that no philosophers had adequately grasped complementarity.

Quantum mechanics seems to be at odds with the classical and common sense picture of the physical world as consisting of causal interactions among known or knowable individual elements. There are at least two different ways in which those classical intuitions are violated by physical reality. One of these, the ineliminability of probabilistic descriptions of physical reality, cuts against intuitions of causal determinism: the view that from a causally complete initial description of a state of affairs, exactly one outcome is certain in itself, even if not certainly known or knowable to us. As unsettling as this ineliminable probabilism may be, the second feature seems to do even more violence to our ordinary ways of thinking about the physical world. The second violation of common sense concerns the observation-independence of events, the view that states of affairs in the physical world *are* some way or another in themselves, logically prior to whatever causal

² In a May 1928 letter to Schrodinger, Einstein wrote: "The soothing philosophy-or religion?-of Heisenberg-Bohr is so cleverly concocted that for the present it offers the believers a soft resting pillow from which they are not easily chased away. Let us therefore let them rest. See Pais, p. 320.

influence is exerted on them through attempts at observation.

At the quantum level there seems to be no such thing as "what actually happens" which could be characterized independently of a particular set of attempts to "measure" such events, where "measuring" is understood as a process of establishing a link between classically-understood observables (e.g., the visible trace of a particle passing through a cloud chamber, etc.) and otherwise unobservable theoretically-implied aspects of physical phenomena. The following sketch describes a single illuminating example, like one with which Bohr says Einstein was particularly impressed³:

Let $AD=AB=BC=DC$. Fire single photons from a source at O to a half-silvered mirror at A.

Theoretically $1/2$ of the photons fired will proceed through A to D, and $1/2$ through A to

B. Place a detector at either D or at B, and particles of the energy of a single photon are detected.

Strange aspect #1: although the probability of detection at either D or B should be .5 (half the time, the photon should have gone the other way), in fact all of the energy of each photon is detected at either point whenever a detector is placed at only D or B. If detectors are placed at both locations, the distribution of detected events becomes .5 at each site.

³ See Bohr, "Discussions with Einstein" in his *Atomic Physics and Human Knowledge* (Wiley & Sons, 1958), p. 50 -51.

Strange aspect #2: If detectors are replaced by mirrors at D and B, interference phenomena can be detected at C, suggesting wave properties and simultaneous ABC and ADC paths of propagation. Yet (again) when the mirror at D (or at B) is replaced with a detector, not 1/2 but all of the energy is detected there (and again, not 1/2 but all the time at either location.)

Now suppose we construct a device which will transform the detector at D into a mirror (a mirrored shutter over the hole where light passes into the detector, say) which completes operation in a quantity of time less than the time required for a photon to travel from A to D. (AD can be as long a distance as we like.) This transformation device can be activated by random number generation, or perhaps even via human intention, if AD is large enough. It seems that an event determining the means of detection and occurring after the photon leaves point A determines whether or not interference phenomena will be detected at C. But that detection-determining event could not be the cause of the directedness of the energy of the photon (whether all along AD, or all along AB, or 1/2 along both routes, or whichever), for the causation would have to have occurred faster than the speed of light, or backward in time. So after the photon passes point A but prior to the time of the occurrence of the detection-determining event, there seems to be no fact of the matter as to along which path the energy is propagating, although after the detection-determining event there would seem to be a true description of how the energy must actually have been propagating along a given path at a given time which could not truly have been ascribed to that energy at that time. Bohr puts the difficulty this way:

"In any attempt of a pictorial representation of the behavior of the photon we

would, thus, meet with the difficulty: to be obliged to say, on the one hand, that the photon always chooses *one* of the two ways and, on the other hand, that it behaves as if it has passed *both* ways." ⁴

Such features of the physical world at the quantum level lead Bohr to refuse to talk about phenomena in isolation from the means used to detect them. "Phenomena" for Bohr were systems linking the implications of quantum theoretical formalism with classically understood observable aspects of reality. This holistic description of phenomena enables the rendering of a logically consistent description of results which are predicted by the logically consistent quantum mechanical formalism and observed to be the case, but which seem logically inconsistent on classical conceptual schemes.

If we adopt a conceptual scheme which makes no reference to how things actually are independently of choices of detection, we avoid contradiction. "If the device at D is set as a detector, then particle-like effects are observed at D, and if the device at D is set as a mirror, then wave-like effects are observed at C." There is no inconsistency in this statement, which takes the phenomena in question as two quantum theoretical/classically-understood observable *wholes*, rather than as one single kind of phenomena detected by two different links to classically-understood observable effects. Bohr claimed that physical reality at the quantum level has simply revealed the limitations of the conceptual scheme which suggests the latter inconsistency-generating sort of description. He proposed a new scheme, one which involves the notion of complementarity. Particle-talk and wave-talk (roughly speaking) are just

4 Bohr, p. 51.

complementary ways, not inconsistent ways, of understanding the physical world.

Complementarity and Mind-Independence Realism

Put thus baldly, Bohr's proposal can be viewed with suspicion, as something of a conceptual trick, an easy way out. It addresses a challenging conceptual problem by eliminating it from the system, rather than by solving it. The situation is of a kind that Thomas Kuhn describes as occurring in generally all scientific revolutions. One mark of what he calls the introduction of new paradigms is that they change what constitutes a scientific problem, meeting resistance from adherents to rival paradigms on just these grounds: the problems seem to be redefined, rather than resolved.

In the case under consideration, it seems that one eliminates the apparent contradictions which arise from viewing (say) photons as mind-independent entities with an inconsistent set of properties by treating the properties of the entities in question as dependent upon our choice of measurement. The contradiction is resolved by making a distinction. There is no single photon with a given set of properties prior to measurement; there are now two photons, in effect, one corresponding to some actual choice of measurement, the other corresponding to some possible but non actual choice of measurement. We want to know what are the real properties of the photon prior to choice of measurement; we are told that we are asking the wrong question.

We can extrapolate beyond the quantum level this complementarist retreat from characterizing the world as having a fixed set of properties independently of our observation of it. Suppose we define the *complementarity thesis* as the claim that *any fully adequate description of the world will contain ineliminable reference to a pair of properties which both apply to the same level of reality, cannot apply simultaneously to the same reality, and whose appropriate predication of that reality is a function the human choice of assessment of that reality.*⁵ Thus stated, the complementarity thesis seems to run afoul of our realist intuition about the world that it has the properties it has independently of our attempts to know which properties those are. We might call any view committed to accommodating this offended intuition a *mind--independence realism*.

Einstein gave expression to his epistemic intuitions about reality in the famous EPR paper. Arguing for the incompleteness of quantum mechanics, he specified what was meant by a system of physical reality: "If, without in any way disturbing a system, we can predict with certainty ... the value of a physical quantity, then there exists an element of physical reality corresponding to this physical quantity."⁶ Bohr's reaction to the EPR effort was to affirm the validity but not the soundness of the argument, claiming that it is precisely their understanding of physical reality which stood in need of amending: "The apparent contradiction in fact discloses only an essential inadequacy of the customary viewpoint of natural philosophy for a rational

⁵ The formulation of this statement of the complementarity thesis is meant to leave it neutral with respect to whether or not fully adequate descriptions and appropriate predications are true.

⁶ See Bohr, p. 58, 59.

account of physical phenomena of the type with which we are concerned in quantum mechanics."⁷

That customary viewpoint of natural philosophy, in Einstein's hands, was a sophisticated mind-independence realism, in which a complete description of physical reality consisted in finding conceptual tools which enabled mathematically certain prediction of all interactions among the set of all mind-independent physical entities. But for all the accompanying mathematical sophistication, Einstein's intuitions about physical reality were not far removed from those of Aristotle, for whom the truth was a kind of relationship between the mind and the external world, a relationship of like formedness. For Einstein, the relationship was something more of behavior correspondence, where the conceptual system predicts a kind of behavior which corresponds to the way the physical reality behaves. Both, nonetheless, are instances of mind-independence realism.

Mind-independence realism has proven to be a difficult world view to pin down, from Aristotle's worries about the pale man in the seventh book of the *Metaphysics* to the modern empiricist search for primary qualities. In an attempt to reconcile classical physics with the skeptical turn taken by the empiricist tradition, Kant retained a realist ontology at the price of introducing an impassable gulf between the phenomenal world--the world as it appears to us through the filters of classical concepts (categories) we can not help but use--and the noumenal world--the world as it is in itself, to which we have no nonfiltered access. Once it became clear that Kantian classical categories are not the only or even the best available set of conceptual lenses,

⁷ Bohr, p. 59.

a new more thoroughgoing skepticism could emerge: earlier and especially in moral philosophy, in the genealogical tradition; later in the philosophy of science, in the work especially of Feyerabend and Kuhn.

Complementarity and Kuhnian Instrumentalism

Although his work is not marked by Nietzschean irony, Kuhn's

The Structure of Scientific Revolutions is nonetheless a genealogy of scientific realism. His epistemology of science is one in which one set of coherent and ever-more powerful conceptual schemes replaces the next, each set enabling a greater measure of formulation and resolution of quantifiable puzzles concerning the physical world. Something like normal science takes place under conditions of communal acceptance of a holistic conceptual scheme--his term is "paradigm"--as promising for the generation of solvable quantifiable puzzles about the physical world; under such conditions

scientific progress amounts to the accumulation of solved scientific puzzles as predicted by the prevailing paradigm. When such schemes come to seem less promising and a rival, potentially more promising paradigm emerges, the nature of scientific activity shifts toward something like articulation and advocacy of the new (or defense of the old) conceptual framework. The process of recession of the old and emergence of the new paradigm is socially complex, and is not guided in any unique direction by the standards of scientific rationality common to all scientific paradigms. Change of prevailing paradigms, or what Kuhn terms "scientific revolutions", only occurs when there is sufficient retention of what are considered solved scientific problems; this is a common requirement of all scientific rationality. Yet what counts as a scientific problem will differ some between paradigms

(recall Bohr's attitude toward the EPR paper); in this regard rival systems have different standards of scientific rationality. There is no rational means for arriving at a unique resolution of conflict between adherents of rival paradigms--common standards of scientific rationality will underdetermine the choice--and thus there can be no non question-begging account of which side has the strongest rational claim in the dispute between adherents of rival paradigms. Thus, while there is scientific progress away from predictively less powerful systems, there is no goal of scientific truth toward which headway might be made. A commitment to the goal of scientific truth is the essence of scientific realism; Kuhn's history means to unmask that commitment.

We tend to think of science as being about a matter of fact, as being about an objective world, because we have achieved scientific progress (as opposed to progress in morality and religion, once but no longer generally hoped for.⁸) But Kuhn argues that we need to reverse our perspective, and see the "facticity" of the physical world not as a material *causal* circumstance enabling scientific consensus and progress, but rather as the *effect* of scientific consensus. Because science is organized in a particular way, it is able to push forward a particular research program, and its highly refined results (defined into highly precise and refinable shape from the outset) constitute scientific progress. Kuhn has turned scientific realism on its head, he thinks.

On a Kuhnian epistemology of science, scientific realism has been left behind, and thus also has mind-independence realism; if there is no truth about

⁸ See the introduction to his Gifford Lectures in MacIntyre, *Three Rival Versions of Moral Inquiry* (Notre Dame, IN: UNDP, 1990).

anything at all, there is no truth about extramental reality. Science does not reveal the nature of the physical world; rather, the conceptual tools we use to predict and control our world are simply elements of the most powerful set yet to emerge, and will likely be superseded by a yet more powerful and incompatible set of conceptual tools.

There is no fact of the matter about which of rival theories is correct; none of the scientific claims made by any rival scientific theories are actually true, and none could be. From the standpoint of Kuhnian epistemology of science, the abandonment of notions of causality and of observation-independent reality in our scientific thinking at the quantum level is simply the latest change in the evolution of scientific paradigms. The Copenhagen interpretation of quantum theory has emerged because and will survive precisely as long as it seems to a sufficiently influential subclass of scientific investigators a more promising direction for formulating solvable quantifiable puzzles than any of its existing rivals. In this way, the instrumentalist finds the advent of complementarity less intellectually jarring than does the mind-independence realist.

Instrumentalism and Craft/Tradition Realism

This comfortable fit between instrumentalism and complementarity seems to scientific realists to come at too high a price. The abandonment of the notions of the mind-independence of extra-mental reality and of scientific truth is indeed a high price. But on the craft/tradition approach to scientific realism, only the former need be left behind; scientific truth can be affirmed as the goal of scientific inquiry without commitment to the mind independence of extra-mental reality. One of Macintyre's great contributions to philosophy has been in showing--to my mind, demonstrating--that

acceptance of the conceptual apparatus which does the work for Kuhnian instrumentalism and for genealogy does not entail the kind of relativism which involves the rejection of an objective standard of truth. MacIntyre's epistemology is a tradition-bound realism.

The heart of MacIntyre's contribution, I think, is not an overly difficult point. MacIntyre acknowledges--rather, has refined its sense and insisted on--the incommensurability of rival conceptual schemes and the role of power in the adjudication of conflicts among adherents to these rivals.⁹ But the Kuhnian variety of instrumentalism in science and the genealogists have failed to adequately take account of the possibility that rival sets of standards inherent in incommensurable conceptual frameworks may well lead to similar conclusions on certain points.¹⁰ It is possible that the conceptual framework inherent in one tradition may prove sufficiently resourceful to persuade adherents of rival traditions toward conversion by using the standards inherent in that rival and soon-to-be-abandoned tradition. Traditions may as a matter of fact prove stronger than their rivals by establishing this sort of asymmetrical relationship of persuasiveness. Persuasive power *is* justification, and it is possible that one tradition may have sufficient justification to overcome any existing rival.¹¹

9 See his "Relativism, Power, and Philosophy" in Kenneth Baynes, et al, ed.s, *Philosophy: End or Transformation?* (Cambridge, MA: The MIT Press, 1987).

10 "What is needed at this point is to consider first more generally the way in which the incommensurability involved in certain radical disagreements may be both recognized and rationally overcome within the context of a certain kind of tradition." *Three Rival Versions of Moral Inquiry* (Notre Dame, IN: UNDP, 1990), p. 118.

11 See my "Freedom and Good in the Thomistic Tradition", *Faith and Philosophy*, Vol. 11, No. 3, June 1994, and "Three Rival Versions of Nonmoral Goodness" in Hancock and Simon, ed.s, *Freedom, Virtue, and the Common Good* (Notre Dame, IN: The American Maritain Association, dist. by UNDP, 1995), p. 160-178.)

It is also possible--for all we know, it is the case with some actually existing tradition--that some tradition, incorporating some scientific account of physical reality, will develop sufficient resources to overcome any rival tradition which human beings are capable of formulating. Such a tradition- call it an *ideal* tradition--might only be known to be the best tradition so far discovered, but it may in fact be the best tradition discoverable. Such a tradition would constitute a mode of interaction between its adherents and the whole of reality, a relationship not only of correspondence nor even only of predictive success, but rather one of all-things-considered stability.

Suppose we roughly define *truth* as *maximal intellectual stability* (taking into account that the desire to understand, to predict, to control, and to benefit from the world are sources of pressure which occasion an exhibition of instability in the intellectual life.) Under such a definition, the judgments, including the scientific judgments, of the members of the ideal tradition would insofar as they were settled all be true, and would constitute the goal of all rational, including scientific, inquiry.

Hence craft/tradition realism builds on Kuhnian instrumentalism and adds to it just the possibility that incommensurability can be rationally overcome without being eliminated; different sets of reasons can point to a common conclusion. In this mere possibility of an ideal tradition, instrumentalism is overcome. For whether or not human beings are such as to be actually able to generate a common intellectually stable perspective toward the world, what Kuhn admits as the minimal common standards of rationality point to that possibility as the goal of scientific inquiry. In the ideal tradition, the maximum units of scientific achievement--the solved quantifiable puzzle- are realized. Any other paradigm with greater puzzle-solving promise would

be a source of intellectual instability to the ideal tradition, a possibility precluded by the definition of an ideal tradition. So while we may be in no position to say whether there *is* an ideal tradition for human beings, we are in a position to see that the ideal tradition and the corresponding notion of truth is the goal of scientific inquiry. Craft/tradition realism overcomes instrumentalism in this way, without reference to the mind-independence of extramental reality.

Complementarity and Craft/Tradition Realism

In overcoming instrumentalism in this way, craft/tradition realism loses none of instrumentalism's ability to accommodate complementarity and the implied loss of the mind-independence of extramental reality. But the craft/tradition model is not only compatible with the falsity of mind independence realism; the craft tradition model implies the falsity of mind independence realism. On the craft/ tradition model, facts about the world (i.e., what can be truly said about it) are those stably imbedded in whichever tradition (if any) can wield persuasive power sufficient for overcoming any other way in which human beings can view the world. Which tradition (if any) is that ideal tradition, and thus which propositions about the world turn out to be true, is a function of the nature of human susceptibility to persuasion. Hence, human nature plays a role in determining which facts are true about the world; there would be no fact of the matter about what would be true of the world, apart from the existence of human beings. All this, on the craft/ tradition model.

That there are no facts about the world at the quantum level apart from the nature of human beings is thus a result indicated by the craft/tradition model of rationality; it is also indicated by complementarity as it applies at the quantum level. For according to it there are no facts about (at least certain features of) reality at the quantum level apart from reference to how that reality will be observed. But all observation consists ultimately in a link between quantum-theoretical formalism and events which must be understood classically and which constitute measurement. Measurement is ineliminably tied to human nature, specifically to human capacities for perception. So there is no fact of the matter about how observation dependent states of physical reality at the quantum level would stand apart from the existence of human beings. Craft/tradition realism and complementarity are thus harmonious in their implications for the mind independence of extramental reality.

Craft/tradition realism thus seems superior to mind-independence realism in accomodating complementarity, and it can make that accomodation without the instrumentalist abandonment of truth as the goal of scientific inquiry.

For the mind-independence realist, the world misbehaves at the quantum level; the classical assumption of mind-independent entities possessing a given set of properties at any given time is falsified by the implications of quantum theory and by predicted and observed experimental results. In the face of such misbehavior, instrumentalism seems an attractive only recourse to scientific realism. But that attractiveness wanes once the craft/tradition notion of an ideal tradition is recognized as coherently characterizable without loss of instrumentalist resources for accomodating complementarity.

On the craft/tradition model, there may be an objective reality the truth of which is discovered by science and other dimensions of human inquiry, but that objective reality is not rightly conceived of as mind-independent, but rather as mind-interdependent: the human mind interacts with extramental reality in a variety of more or less stable ways, and the goal of human inquiry is to discover the most stable ways, and perhaps the most stable way, embodied in the ideal tradition. The relative stability of such modes of interaction seems not to be a product of whim or choice, but of the way things are, of the way the human mind and the whole reality of which it is part are. Human inquiry aims at stable understanding of the sort Plato described as captured in an account which can stand every test of being and emerge with the argument still on its feet.¹² Whether or to what extent we succeed in achieving such maximal stability remains an untold story, but the craft/tradition model seems likely to contribute to that stability through the the early chapters on complementarity.

¹² *Republic* 534c